# Week 8 Report: Stacking Ensemble Variations and Evaluation

## 1. Modeling Approach

This week we built a stacking ensemble that combines the best models from Weeks 6 and 7, and evaluated three variations of it on the same 27-feature reduced set prepared in Week 5. In stacking, several base models each predict a probability of response, and a second model, the meta-model, learns how much weight to give each one. We chose this approach for three reasons:

- **A more flexible single model is unlikely to help.** Week 7 showed that a larger neural network memorized the training customers (training PR-AUC 1.000, validation 0.678), and Week 6 showed the same for the tree ensembles. Adding flexibility to one model has so far added noise, not signal.
- **Combining models can help if they make different mistakes.** Logistic regression, gradient boosting, and the neural network learn in different ways. If they rank some customers differently, a meta-model can combine their strengths.
- **It builds directly on earlier work.** Each base model uses the best settings found in Weeks 6 and 7, so the ensemble tests whether the strongest version of each model type adds value when combined.

Three base models were used:

| Base model | Source | Settings | Validation PR-AUC |
|---|---|---|---|
| Logistic regression | Week 6, best model so far | C = 3, L2 penalty, no class weights | 0.732 |
| Gradient boosting | Week 6, best tree model | 200 trees, learning rate 0.05, max depth 2, subsample 0.8 | 0.693 |
| Neural network | Week 7, Variation 1 | One hidden layer of 8 units, alpha = 3.0 | 0.725 |

Gradient boosting was used rather than the random forest because it scored higher in Week 6 (0.693 vs 0.650).

**Validation design.** As in Weeks 6 and 7, each model was trained on four-fifths of the training set and scored on the remaining fifth, using 5-fold stratified cross-validation repeated 3 times, for 15 train/validation splits. These are the same 15 splits as Weeks 6 and 7, so results can be compared fold by fold. Training metrics are measured on the data each model learned from; validation metrics are measured on customers the model did not see. The test set was not used this week and is kept for the final evaluation.

The meta-model must not learn from predictions that base models made on customers they were trained on, because those predictions are overly confident. Inside each training fold, the base models' predictions are therefore produced with an inner 5-fold split, so each customer's prediction comes from base models that did not see that customer. The validation fold is never used to train either the base models or the meta-model.

**Do the base models make different mistakes?** Stacking can only beat its best base model if the base models disagree on some customers, so we checked this before comparing the variations. Logistic regression and the neural network rank customers almost identically: their rankings have a Spearman correlation of 0.975, and 91.5% of the 270 customers each would contact first are the same. Gradient boosting differs more: its correlation with logistic regression is 0.877, and the two share 76.3% of their top 270. The neural network therefore adds little that logistic regression does not already capture, and gradient boosting is the only base model with room to add new information. The supporting tables are in Appendix A.

## 2. Model Complexity

The stacking ensemble is the most complex approach so far, because it contains the previous models rather than replacing them. Its complexity comes from three sources:

- **More models.** Variation 1 combines two base models, and Variations 2 and 3 combine three. Every variation includes the Week 7 neural network (233 weights), and Variations 2 and 3 also include 200 boosted trees.
- **More training.** The inner split means each base model is trained 6 times per training fold: 5 times to produce the meta-model's inputs, and once on the full training fold. Variation 1 therefore trains 12 base models per fold, and Variations 2 and 3 train 18.
- **A second learning layer.** The meta-model learns its own weights on top of the base models. In Variations 1 and 2 it has only 2 or 3 inputs. In Variation 3 it also receives the 27 original features, giving it 30 inputs.

The added complexity makes the ensemble the hardest model so far to explain. A prediction depends on several models plus the weights that combine them, so no single coefficient shows how a feature raises or lowers the chance of response. With only 234 responders, the main risk is the same as in earlier weeks: the more freedom the meta-model has, especially in Variation 3, the more likely it is to fit noise.

## 3. Hyperparameters Evaluated

Each variation changes at least one hyperparameter of the stacking model:

| Hyperparameter | Variation 1 | Variation 2 | Variation 3 | What it controls and why we varied it |
|---|---|---|---|---|
| Base models (estimators) | LR + NN | LR + GB + NN | LR + GB + NN | Which models are combined. Variation 1 tests whether the two best single models add anything together. Variation 2 adds gradient boosting, which uses decision trees and ranks customers most differently. |
| passthrough | False | False | True | Whether the meta-model also sees the 27 original features. This lets it correct the base models directly, but gives it far more room to overfit. |

| Hyperparameter | Variation 1 | Variation 2 | Variation 3 | What it controls and why we varied it |
|---|---|---|---|---|
| Meta-model C | 1.0 | 1.0 | 0.1 | Regularization of the meta-model; smaller values mean a stronger penalty. Variation 3 gets a stronger penalty to offset its extra inputs. |
| Meta-model inputs | 2 | 3 | 30 | The result of the settings above, shown to compare complexity. |

All other settings were held fixed so the comparison isolates these choices: base model settings from Weeks 6 and 7, an inner 5-fold stratified split (seed 42), positive-class probabilities as meta-model inputs, a logistic regression meta-model, and a standard scaler before the meta-model so its inputs are on the same scale. No class weights or SMOTE were used, consistent with Weeks 6 and 7.

## 4. Performance Metrics

Only 14.6% of training customers responded. A model that predicts "no response" for everyone would be 85.4% accurate while finding no responders, so accuracy is misleading here. We used the same five metrics as Weeks 6 and 7, so the ensemble can be compared directly with the earlier models:

- **PR-AUC (average precision)**, our main metric. It measures how well the model ranks responders above non-responders, with emphasis on the minority class. Random guessing scores 0.146, the response rate. This matches the business use: the model's job is to rank customers so the company contacts the most promising ones first.
- **ROC-AUC.** The probability that a randomly chosen responder is ranked above a randomly chosen non-responder. With imbalanced classes it can look high even when precision is poor, so we treat it as secondary.
- **Precision.** Of the customers the model flags, the share who actually respond. Low precision means wasted contact costs.
- **Recall.** Of all responders, the share the model flags. Low recall means missed revenue.
- **F1 score.** The harmonic mean of precision and recall, summarizing the trade-off between them in one number.

PR-AUC and ROC-AUC use the predicted probabilities directly. Precision, recall, and F1 use a cutoff of 0.5 to turn probabilities into contact decisions. Because a fixed cutoff does not treat every model's probabilities equally, we rely on PR-AUC for the main comparison.

## 5. Results: Training and Validation Metrics

Variation 2 scored highest on validation PR-AUC (0.735), ahead of Variation 1 (0.731) and Variation 3 (0.726). All values are averages over the 15 cross-validation folds; precision, recall, and F1 use a 0.5 cutoff. Bold marks the best validation value for each metric among the three variations.

| Model | Dataset | PR-AUC | ROC-AUC | Precision | Recall | F1 |
|---|---|---|---|---|---|---|
| Variation 1 (LR + NN) | Training | 0.779 | 0.943 | 0.795 | 0.579 | 0.670 |
| Variation 1 (LR + NN) | Validation | 0.731 | 0.929 | 0.744 | **0.548** | **0.629** |
| Variation 2 (LR + GB + NN) | Training | 0.813 | 0.953 | 0.831 | 0.599 | 0.696 |
| Variation 2 (LR + GB + NN) | Validation | **0.735** | **0.931** | 0.759 | 0.535 | 0.625 |
| Variation 3 (LR + GB + NN, passthrough, C = 0.1) | Training | 0.799 | 0.948 | 0.837 | 0.585 | 0.688 |
| Variation 3 (LR + GB + NN, passthrough, C = 0.1) | Validation | 0.726 | 0.929 | **0.771** | 0.521 | 0.619 |

Reference points (same 15 folds):

| Reference | Dataset | PR-AUC | ROC-AUC | Precision | Recall | F1 |
|---|---|---|---|---|---|---|
| Logistic regression (tuned, Week 6) | Training | 0.772 | 0.941 | 0.784 | 0.576 | 0.664 |
| Logistic regression (tuned, Week 6) | Validation | 0.732 | 0.930 | 0.740 | 0.558 | 0.633 |
| Prior-campaign rule | Validation | 0.395 | - | - | - | - |
| Random guessing | Validation | 0.146 | 0.500 | - | - | - |

## 6. How Training and Validation Metrics Change Across Variations

The gap between training and validation PR-AUC shows how much each variation overfits:

| Model | Training PR-AUC | Validation PR-AUC | Gap | Validation SD across folds | Folds beating tuned LR (of 15) |
|---|---|---|---|---|---|
| Variation 1 (LR + NN) | 0.779 | 0.731 | 0.048 | 0.062 | 7 |
| Variation 2 (LR + GB + NN) | 0.813 | 0.735 | 0.079 | 0.060 | 8 |
| Variation 3 (passthrough, C = 0.1) | 0.799 | 0.726 | 0.073 | 0.064 | 5 |
| Logistic regression (tuned, Week 6) | 0.772 | 0.732 | 0.040 | 0.068 | - |

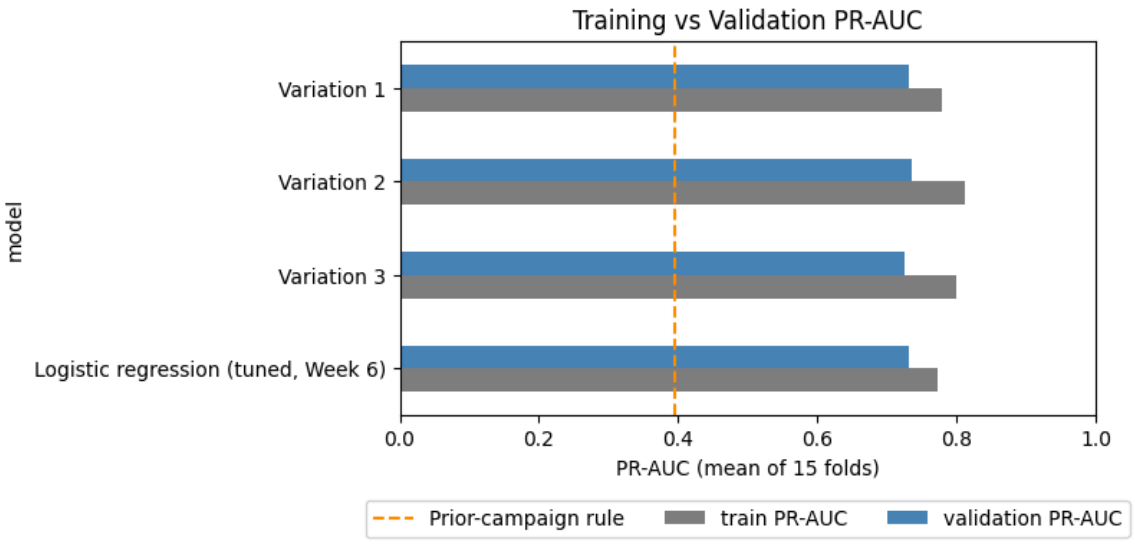



**Figure 1. Training vs validation PR-AUC by model.** The dashed line is the prior-campaign rule (0.395). The training and validation bars are close for every model, showing the small gaps in the table above.

- **Adding gradient boosting raised both scores.** Moving from Variation 1 to Variation 2 raised training PR-AUC from 0.779 to 0.813 and validation PR-AUC from 0.731 to 0.735. Gradient boosting is the base model that ranks customers most differently (Section 1), so it is the only one that gives the meta-model new information to combine.
- **Passing the original features through did not help.** Variation 3 scored lower than Variation 2 on both training (0.799) and validation (0.726). The stronger penalty kept it from memorizing, but the 27 features are already used by every base model, so giving them to the meta-model again added inputs without adding information.
- **Overfitting stayed small in every variation.** The gaps were 0.048, 0.079, and 0.073, close to logistic regression's 0.040 and far below Week 7's Variation 3 (0.322) and Week 6's random forest (0.318). Gradient boosting alone overfit in Week 6 (training 0.885, validation 0.693), but inside the ensemble its influence is limited by the meta-model, which is trained only on predictions for customers the base models did not see.
- **Precision and recall trade off at the 0.5 cutoff.** Validation precision rises from 0.744 to 0.759 to 0.771 while recall falls from 0.548 to 0.535 to 0.521. Validation F1 is highest for Variation 1 (0.629), but all three are below the tuned logistic regression (0.633), and F1 depends on the cutoff, whereas PR-AUC does not.
- **Fold-to-fold variation is larger than the differences.** Validation PR-AUC varies by about 0.06 across folds, far more than the 0.009 spread between the three variations. Variation 2 beat Variation 1 on 10 of 15 folds, with a mean difference of +0.003 and a fold-to-fold standard deviation of the difference of 0.009, so its lead is consistent but small.

In short, combining models kept overfitting low, unlike the larger single models in Weeks 6 and 7, but it added very little validation performance.

## 7. Validation Comparison and Best Model

On the validation folds, Variation 2 scored highest on PR-AUC (0.735) and ROC-AUC (0.931). Variation 1 was second on PR-AUC (0.731) and had the highest recall (0.548) and F1 (0.629). Variation 3 was last on PR-AUC (0.726), although it had the highest precision (0.771).

**Best model: Variation 2 (logistic regression, gradient boosting, and neural network; meta-model C = 1.0; no passthrough).** We selected it for three reasons:

1. Best validation performance on the main metric (PR-AUC 0.735), and on ROC-AUC (0.931).
2. Consistent lead over the runner-up. It beat Variation 1 on 10 of 15 folds, which shows that adding gradient boosting helped on most splits rather than on a few lucky ones.
3. Moderate overfitting. Its train-validation gap of 0.079 is larger than Variation 1's (0.048) but small compared with any single flexible model so far, so its performance should hold for new customers.

**Comparison with earlier weeks.** Variation 2 has the highest validation PR-AUC of any model so far, but its lead over the tuned logistic regression from Week 6 (0.732) is within noise. The mean difference is +0.002, it beat logistic regression on only 8 of 15 folds, and the fold-to-fold standard deviation of the difference is 0.017, more than eight times the gain. Variation 1 beat logistic regression on 7 of 15 folds and Variation 3 on 5. The ensemble therefore matches logistic regression rather than improving on it.

The base model check in Section 1 explains why. The neural network ranks customers almost the same way as logistic regression, so it adds nothing new. Gradient boosting ranks customers differently, but it is also the weakest base model (0.693), so what it adds is partly noise. With only 234 responders, and with the main interactions already captured by the Week 5 features, there is little signal left that a combination of models can find. All three variations still beat the prior-campaign rule (0.395) by a wide margin, Variation 2 by 0.340.

## 8. Conclusion

Variation 2, which combines logistic regression, gradient boosting, and the neural network, is the best model this week. It had the highest validation PR-AUC (0.735) and ROC-AUC (0.931), and it beat Variation 1 on 10 of 15 folds. Passing the original features to the meta-model (Variation 3) did not help.

Unlike the larger single models in Weeks 6 and 7, the ensembles did not memorize the training data: every variation's train-validation gap stayed below 0.08. However, the best ensemble is effectively tied with the tuned logistic regression from Week 6 (+0.002, better on 8 of 15 folds) while training 18 base models per fold and being much harder to explain. The test set was not used this week and remains reserved for the final evaluation.

## Appendix A: Do the Base Models Make Different Mistakes?

Each customer was scored by each base model trained without that customer (one pass of 5-fold cross-validated predictions), and the resulting rankings were compared. Because this uses one pass of 5 folds, the PR-AUC values in Table A1 differ slightly from the 15-fold averages in Section 1.

**Table A1. PR-AUC of each base model (cross-validated predictions)**

| Base model | PR-AUC |
|---|---|
| Logistic regression | 0.724 |
| Gradient boosting | 0.695 |
| Neural network | 0.721 |

**Table A2. Spearman correlation between base model rankings** (1.0 means identical rankings)

| | Logistic regression | Gradient boosting | Neural network |
|---|---|---|---|
| Logistic regression | 1.000 | 0.877 | 0.975 |
| Gradient boosting | 0.877 | 1.000 | 0.896 |
| Neural network | 0.975 | 0.896 | 1.000 |

**Table A3. Share of the top 270 customers two models have in common** (270 is the number flagged at the Week 6 cutoff)

| Pair | Shared top 270 |
|---|---|
| Logistic regression vs Neural network | 91.5% |
| Gradient boosting vs Neural network | 78.1% |
| Logistic regression vs Gradient boosting | 76.3% |